PROMPT FRAMEWORK DESIGN

CRISP, RTF, Templates, Reusability, and Practical Experimental Evaluation

Academic Research & Practical Experimental Study

Core Framework

CRISP → RTF → Reusable Templates → Experimentation → Evaluation → Error Analysis → Refinement → Versioning

Prepared as a structured prompt-engineering research assignment

Abstract.

Prompt engineering has become an important component of modern Artificial Intelligence (AI) systems because the quality, consistency, and usefulness of Large Language Model (LLM) outputs are strongly influenced by how instructions are structured. This study investigates structured prompting through the **CRISP framework, RTF framework, prompt templates, and prompt reusability**. The research develops a reusable hybrid prompt framework and evaluates it through a practical experiment. The experiment compares different prompting approaches on identical tasks and evaluates outputs using task accuracy, relevance, completeness, format compliance, consistency, and instruction-following. A controlled methodology keeps the AI model, task, and input information constant while changing prompt structure. The study compares an unstructured prompt, RTF prompt, CRISP prompt, and reusable CRISP-RTF template. The objective is to investigate whether structured prompt engineering improves reliability and repeatability rather than simply assuming that longer prompts are better.

1. Introduction

Generative AI and Large Language Models have changed how users interact with computer systems. Instead of programming every behavior explicitly, users can communicate with AI systems using natural-language instructions. However, natural-language interaction introduces an engineering problem: **How should instructions be designed so that an AI system consistently produces the intended output?** A simple request such as “Explain Artificial Intelligence” does not define the audience, scope, depth, structure, terminology, or quality requirements. Structured prompts provide these requirements explicitly. Prompt engineering therefore involves designing instructions that establish context, role, objective, task, constraints, input, output requirements, and verification criteria.

2. Problem Statement

Modern LLMs can generate high-quality responses, but outputs may vary when prompts are vague, incomplete, or inconsistently structured. Common problems include ambiguous instructions, incomplete answers, irrelevant information, inconsistent formatting, incorrect assumptions, failure to follow constraints, and variation between repeated executions. Users also frequently create prompts manually for individual tasks, making the approach difficult to scale. Therefore, a structured and reusable prompt framework is needed to organize instructions, reduce ambiguity, improve consistency, support reuse, enable evaluation, and support repeatable experimentation.

3. Research Aim

To design and experimentally evaluate a reusable prompt framework based on CRISP, RTF, templates, and structured prompt engineering.

4. Research Objectives

• Examine the principles of prompt engineering.

• Analyze the CRISP framework.

• Analyze the RTF framework.

• Compare structured and unstructured prompts.

• Develop a CRISP-RTF hybrid framework.

• Design reusable prompt templates.

• Conduct a controlled practical experiment.

• Measure the quality of AI outputs.

• Analyze errors and inconsistencies.

• Evaluate the usefulness of prompt reusability.

• Identify limitations of structured prompting.

• Propose recommendations for future AI systems.

5. Research Questions

RQ1. Does structured prompt engineering improve the quality of LLM outputs compared with an unstructured prompt?

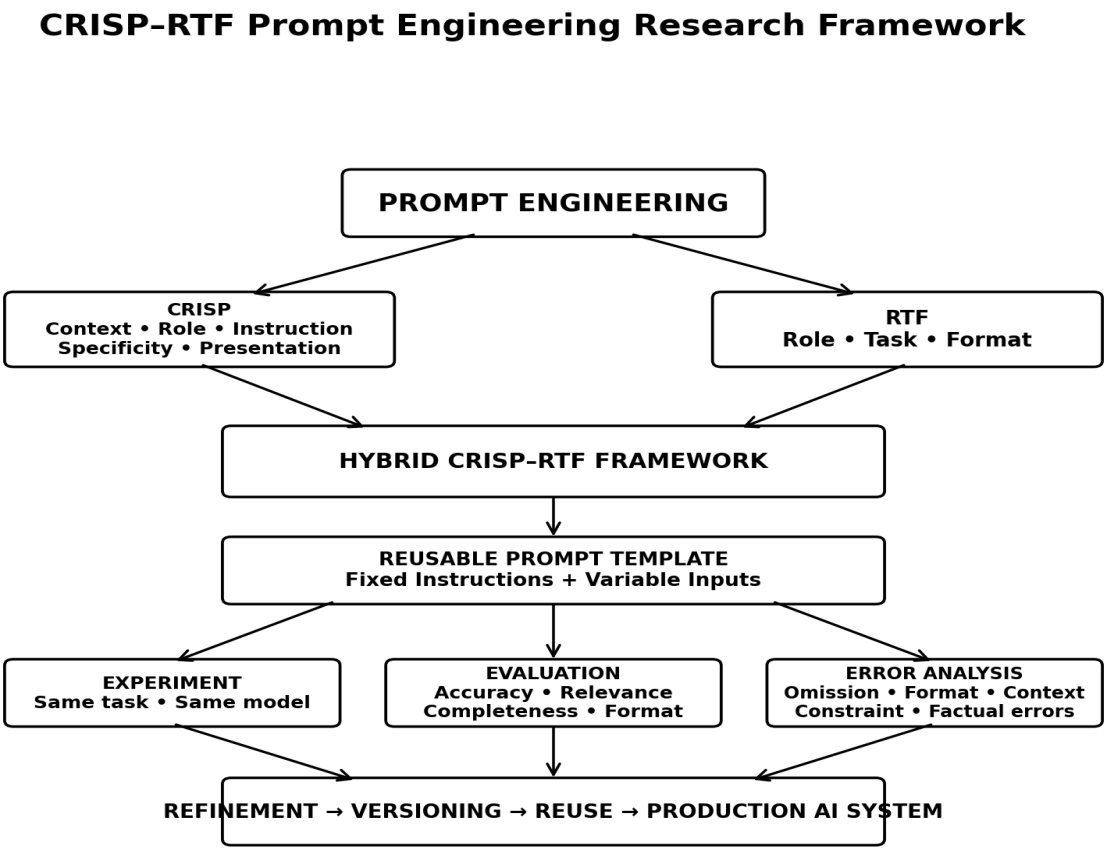
RQ2. How does RTF prompting perform compared with CRISP prompting for structured AI tasks?

RQ3. Does the CRISP-RTF hybrid framework improve instruction-following and output-format compliance?

RQ4. Does the use of reusable prompt templates improve consistency across multiple similar tasks?

RQ5. Which evaluation dimensions are most affected by structured prompting?

RQ6. What types of errors remain even when structured prompt frameworks are used?



8. Methodology

Research Design: Controlled experimental design. The experiment changes prompt structure while keeping other important variables as constant as practical.

Independent Variable: Prompting method — Unstructured, RTF, CRISP, and CRISP-RTF reusable template.

Dependent Variables: Accuracy, relevance, completeness, instruction following, format compliance, constraint adherence, and consistency.

Controlled Variables: AI model, task, input information, temperature/generation settings where available, output language, evaluation rubric, and number of test cases.

9. Experimental Groups

Group	Prompt Type	Description
G1	Unstructured	Basic natural-language request
G2	RTF	Role + Task + Format
G3	CRISP	Context + Role + Instruction + Specificity + Presentation
G4	CRISP-RTF	Hybrid reusable structured template

10. Experimental Task

Explain Artificial Intelligence to undergraduate students who understand basic programming but have limited knowledge of AI. The answer must cover: definition of AI, Machine Learning, Deep Learning, Generative AI, applications, benefits, limitations, and one practical example. The expected output should contain clear headings and approximately 700–900 words.

11. Experimental Prompt 1 — Unstructured

```
Explain Artificial Intelligence to undergraduate students.
Include Machine Learning, Deep Learning, Generative AI,
applications, benefits, limitations, and examples.
```

12. Experimental Prompt 2 — RTF

```
ROLE:
Act as an Artificial Intelligence professor.

TASK:
Explain Artificial Intelligence to undergraduate students
who understand basic programming but have limited AI knowledge.
Cover AI, Machine Learning, Deep Learning, Generative AI,
applications, benefits, limitations, and practical examples.

FORMAT:
Use clear headings and approximately 700–900 words.
```

13. Experimental Prompt 3 — CRISP

```
CONTEXT:
The audience consists of undergraduate computer science students
with basic programming knowledge but limited understanding of AI.
```

```
ROLE:
Act as a university professor specializing in Artificial Intelligence.

INSTRUCTION:
Explain Artificial Intelligence and introduce the major concepts
required for understanding modern AI.

SPECIFICITY:
Include AI definition, Machine Learning, Deep Learning,
Generative AI, applications, benefits, limitations, and practical examples.
Use technically accurate terminology and explain advanced terms
when they first appear.

PRESENTATION:
Use clear headings, logical progression, and approximately 700–900 words.
```

14. Experimental Prompt 4 — CRISP-RTF Reusable Template

```
ROLE:
Act as a {{EXPERT_ROLE}}.

CONTEXT:
The audience is {{AUDIENCE}}.
Their existing knowledge level is {{KNOWLEDGE_LEVEL}}.

OBJECTIVE:
Explain {{TOPIC}} clearly and accurately.

TASK:
Cover the following concepts:
{{REQUIRED_CONCEPTS}}

SPECIFIC REQUIREMENTS:
- Use technically accurate terminology.
- Explain specialized terminology.
- Include practical examples.
- Maintain logical progression.
- Avoid unnecessary repetition.

CONSTRAINTS:
- Target length: {{WORD_RANGE}}
- Do not assume advanced prior knowledge.
- Do not introduce unsupported claims.

FORMAT:
1. Introduction
2. Core Concepts
3. Practical Examples
4. Applications
5. Benefits
6. Limitations
7. Conclusion

QUALITY CONTROL:
Verify that every required concept has been addressed
and that the requested structure has been followed.
```

15. Test Cases

Test Case	Topic	Complexity
TC1	Artificial Intelligence	Medium
TC2	Machine Learning	Medium
TC3	Generative AI	Medium
TC4	Prompt Engineering	High

Test Case	Topic	Complexity
TC5	RAG Architecture	High

Experimental volume: 4 prompt methods × 5 tasks = 20 runs. For stronger reliability, 3 runs per condition gives 60 outputs.

16. Evaluation Rubric

Each output is evaluated on a 1–5 scale: 1 = Very Poor, 2 = Poor, 3 = Acceptable, 4 = Good, 5 = Excellent. Dimensions: accuracy, relevance, completeness, instruction following, format compliance, constraint adherence, and consistency.

17. Evaluation Matrix

Method	Accuracy	Relevance	Completeness	Instruction	Format	Consistency
Unstructured	/5	/5	/5	/5	/5	/5
RTF	/5	/5	/5	/5	/5	/5
CRISP	/5	/5	/5	/5	/5	/5
CRISP-RTF	/5	/5	/5	/5	/5	/5

Actual scores must be assigned after observing the model outputs; they should not be invented in an academic report.

18. Practical Comparison Experiment

The central experiment uses the same task, model, input, and evaluation criteria while changing the prompt structure. Outputs are collected, scored, compared quantitatively, and analyzed for errors. The key principle is to change prompting structure as much as reasonably possible while keeping other experimental variables constant.

19. Observed-Result Protocol

A rigorous results section must distinguish **observed results** from **expected patterns**. The experiment may reveal broad responses, omissions, or less predictable organization in the unstructured condition; clearer task and format alignment in RTF; stronger contextual grounding and requirement handling in CRISP; and standardized variable-driven behavior in CRISP-RTF. These are hypotheses/observation categories until the actual outputs are scored.

20. Quantitative Analysis

Mean Score = Σ Scores / Number of Evaluations. A six-dimension overall score can be calculated as: (Accuracy + Relevance + Completeness + Instruction Following + Format + Consistency) / 6. For multiple test cases, the method mean is the sum of test-case scores divided by the number of test cases.

21. Consistency Measurement

If the same prompt is executed repeatedly, variation can be inspected using Range = Maximum Score – Minimum Score. Lower variation indicates more stable performance under that evaluation setup, although standard deviation or other statistical measures can provide a stronger analysis.

22. Error Analysis

• Omission — required concept missing.

• Format Failure — requested structure not followed.

• Context Failure — response inappropriate for intended audience.

• Constraint Failure — word count, scope, or another requirement violated.

• Relevance Failure — unnecessary or unrelated content.

• Factual Error — incorrect information.

• Instruction Conflict — incompatible instructions within the prompt.

23. Error Analysis Table

Error	Unstructured	RTF	CRISP	CRISP-RTF
Missing topic	Record	Record	Record	Record
Format violation	Record	Record	Record	Record
Audience mismatch	Record	Record	Record	Record
Constraint violation	Record	Record	Record	Record
Irrelevant content	Record	Record	Record	Record

24. Prompt Reusability Experiment

The same template structure can be applied to Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, and Prompt Engineering by changing only variables such as role, audience, topic, task, constraints, and output format. Evaluation should measure output consistency, formatting consistency, completeness, adaptation, and the amount of manual modification required.

25. Reusability Metrics

Reuse Efficiency = Successful Tasks / Template Adaptations. Prompt Modification Rate = (Number of Modified Instructions / Total Prompt Components) × 100. These metrics provide a practical way to describe how much adaptation a template requires.

26. Methodology Summary

Component	Description
Research Type	Experimental
Approach	Quantitative + qualitative
Independent Variable	Prompt framework
Dependent Variables	Output quality metrics
Prompt Conditions	Unstructured, RTF, CRISP, CRISP-RTF
Test Cases	Multiple AI tasks
Evaluation	1–5 rubric
Repetition	Multiple runs

Component	Description
Analysis	Mean, variation, error analysis
Secondary Experiment	Template reusability

27. Validity and Reliability

• Same Model — use the same model for all conditions.

• Same Input — provide equivalent task information.

• Same Evaluation Criteria — use one standardized rubric.

• Multiple Test Cases — avoid conclusions from a single task.

• Multiple Runs — repeat conditions where possible.

• Blind Evaluation — if possible, evaluators should not know which prompt produced an output.

28. Threats to Validity

• Model variation

• Sampling variation

• Evaluator subjectivity

• Prompt length/token effects

• Task dependency

Results should not automatically be generalized to every AI system, model, or task.

29. Limitations

• Limited number of models may be used.

• The number of test cases may be small.

• Human evaluation can introduce subjectivity.

• LLM behavior can change across model versions.

• Prompt performance depends on task complexity.

• Longer prompts may increase token usage.

• Structured prompting cannot eliminate hallucinations.

• A prompt framework cannot compensate for missing or unreliable source information.

30. Discussion

Prompt complexity is not the same as prompt quality. Adding more instructions does not automatically improve AI performance. Effective prompts should contain relevant structure. CRISP contributes contextual and constraint-oriented structure, while RTF provides a compact Role–Task–Format foundation. Templates introduce separation between stable instructions and variable data, similar to reusable functions in software engineering.

31. Advanced Insight: Prompt Engineering as Software Engineering

Traditional prompting can be represented as User → Prompt → AI → Output. Production prompt engineering extends this to Prompt Specification → Template → Variables → Model → Output → Evaluation → Logging → Versioning → Optimization. A production prompt should ideally have an identifier, version, purpose, inputs, constraints, output schema, evaluation criteria, test cases, and revision history.

32. Proposed Prompt Lifecycle

Requirement → Prompt Design → CRISP / RTF → Template → Test → Evaluate → Error Analysis → Refine → Version → Deploy → Monitor → Improve

33. Practical Recommendations

• For simple tasks: use RTF.

• For context-heavy tasks: use CRISP.

• For repeated tasks: use a Prompt Template.

• For production AI systems: use CRISP-RTF + Templates + Evaluation + Versioning.

• For AI agents: add tool instructions, state management, verification, and structured outputs.

34. Final Proposed Framework

CRISP-RTF-P Framework

• C — Context: define the situation.

• R — Role: define the AI's role.

• I — Instruction: define the required action.

• S — Specificity: define requirements and constraints.

• P — Presentation: define the output structure.

• RTF Layer: explicit Role, Task, Format.

• Template Layer: convert prompts into reusable variables.

• Evaluation Layer: measure output quality.

• Iteration Layer: refine based on observed errors.

Prompt System = CRISP + RTF + Template + Evaluation + Iteration

35. Conclusion

This research developed a structured prompt framework combining CRISP, RTF, prompt templates, and reusable prompt engineering. The research questions investigate whether structured prompting improves AI output quality, how CRISP and RTF compare, whether their combination provides additional benefits, and whether reusable templates improve consistency. A controlled practical experiment compares Unstructured, RTF, CRISP, and CRISP-RTF conditions using the same tasks, inputs, model, and evaluation criteria.

Outputs can be evaluated for accuracy, relevance, completeness, instruction following, format compliance, constraint adherence, and consistency. The methodology also introduces error analysis and a dedicated template-reusability experiment. The central research issue is not whether longer prompts are automatically better, but whether well-structured, reusable, testable prompts provide more reliable AI behavior than unstructured instructions.

36. Research Contribution

The main contribution is the proposed **CRISP-RTF-P Prompt System**, combining CRISP, RTF, prompt templates, reusability, experimental evaluation, error analysis, and versioning. Rather than treating a prompt as a single instruction, the framework treats it as a reusable and measurable component of an AI system.

37. Complete Master Prompt Template

```
================================================
        CRISP-RTF-P MASTER PROMPT
================================================

[ROLE]
Act as a {{ROLE}} specializing in {{DOMAIN}}.

[CONTEXT]
Project: {{PROJECT}}
Audience: {{AUDIENCE}}
Existing Knowledge: {{KNOWLEDGE_LEVEL}}
Background: {{CONTEXT}}

[OBJECTIVE]
The desired outcome is:
{{OBJECTIVE}}

[TASK]
Perform the following task:
{{TASK}}

[REQUIREMENTS]
1. {{REQUIREMENT_1}}
2. {{REQUIREMENT_2}}
3. {{REQUIREMENT_3}}

[CONSTRAINTS]
- {{CONSTRAINT_1}}
- {{CONSTRAINT_2}}
- {{CONSTRAINT_3}}

[INPUT]
{{INPUT}}

[PROCESS]
1. Understand the context.
2. Identify the objective.
3. Analyze the input.
4. Execute the task.
5. Verify the result.
6. Produce the final response.

[OUTPUT FORMAT]
{{OUTPUT_FORMAT}}

[QUALITY CONTROL]
Verify accuracy, relevance, completeness, instruction following,
constraint adherence, format compliance, and consistency.
If critical information is missing, identify it explicitly.
Do not fabricate unsupported information.
```

```
[VERSION]
Prompt Version: {{VERSION}}

[END]
```

38. Final Research Model

CRISP + RTF → Hybrid Framework → Prompt Template → Reusability → Experimentation → Evaluation → Error Analysis → Refinement → Versioning → Production AI System

> **Core Principle:** A high-quality prompt is not merely a well-written instruction; it is a structured, testable, reusable, and measurable interface between human intent and AI behavior.